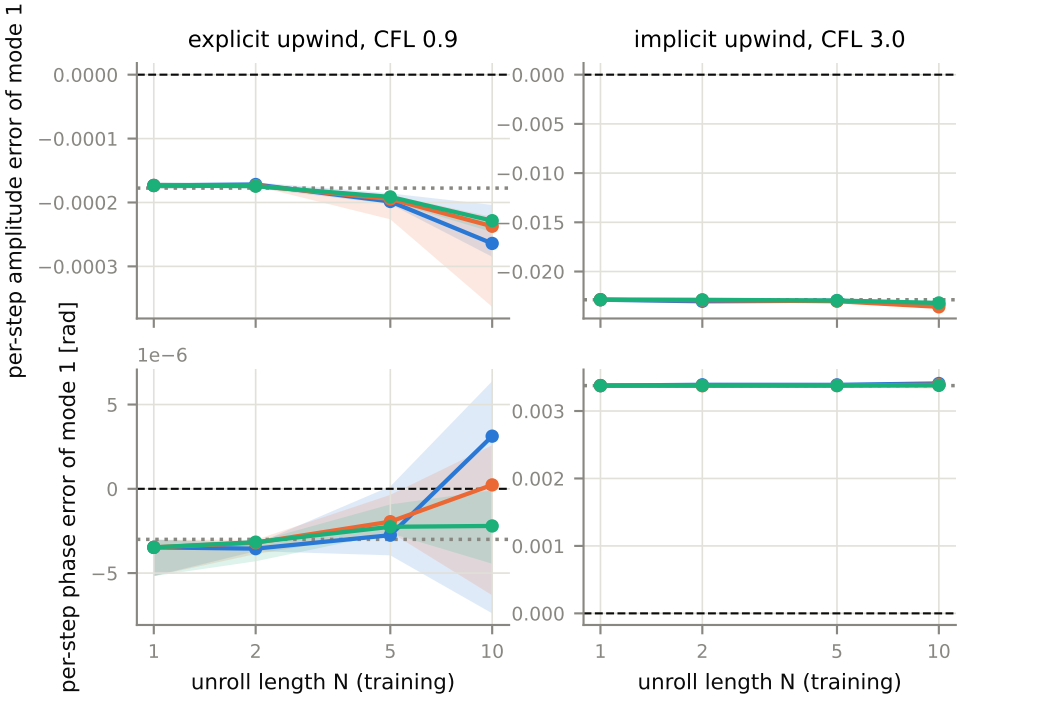
FNO: learned one-step Fourier symbol vs. exact and vs. training solver
explicit upwind, CFL 0.9
implicit upwind, CFL 3.0
per-step amplitude error of mode 1
per-step phase error of mode 1 [rad]
1e−6
0.0000
−0.0001
−0.0002
−0.0003
0.000
−0.005
−0.010
−0.015
−0.020
5
0
−5
0.003
0.002
0.001
0.000
1
2
5
10
unroll length N (training)
low-fidelity solver
exact
data-unrolled (sup)
solver-in-loop, no grad (div-NOG)
solver-in-loop, with grad (div-WIG)